A LITTLEWOOD–RICHARDSON RULE FOR FOREST POLYNOMIALS VIA THE SCHUBERT BIALGEBRA

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Abstract.

1. Introduction

2. Set-valued inversions tableaux

Definition 2.0.1. Let w be a permutation such that $\mathfrak{m}(w) \leq n$. Suppose $T : I(w) \to \mathbf{2}^{[n]} \setminus \{\varnothing\}$ is a function mapping a root to a nonempty subset of $[n]$. Define relations $\leq$ and $<$ on these sets by declaring that $A \leq B$ if $\max(A) \leq \min(B)$ and $A < B$ if $\max(A) < \min(B)$. We say that T is a *set-valued inversions tableau* if the following conditions are satisfied:

- If $i < j < k$ and $(i,j), (j,k), (i,k) \in I(w)$, then either $T(i,j) \leq T(i,k) \leq T(j,k)$ or $T(j,k) \leq T(i,k) \leq T(i,j)$.
- If $i > 0$ and $j < k$ are such that $(i,j), (i,k) \in I(w)$, then $T(i,j) < T(i,k)$.
- $T(i, i+1) \leq \{i\}$ for all i.

For $(i,j) \in I(w)$, define $\lambda_T(i,j)$ as follows. Suppose $T(i,j) = r$. Let (i,k) be the root such that $T(i,k) = r$ and k is minimal. Then $\lambda_T(i,j) = i + j + 1 - k$.

Theorem 2.1. *Suppose w is a permutation. Then we have the following formula for the double Grothendieck polynomial of w:*

$$\mathfrak{G}_w^{(\beta)}(x;y) = \sum_T \prod_{(i,j)\in I(w)} \beta^{|T(i,j)|-1} \prod_{r\in T(i,j)} x_r \oplus y_{\lambda_T(i,j)-r}$$

where the sum is over all set-valued inversions tableaux of w.

Definition 2.0.2. An *staircase word graph* is simply a subset of $\mathbb{N}^2$. We impose the same ordering on these as RC graphs. That is, we say that $(a,b) \leq (c,d)$ if $a < c$ or $a = c$ and $b > d$. We define $\mathbf{word}(G)$ to be the word obtained by reading the elements $i + j - 1$ in the linear order given by $\leq$. We then say that w_G is the Demazure product of $\mathbf{word}(G)$.

For a set-valued inversions tableau T, we can define a staircase word graph G as follows. If w is the identity, then G is the empty set. Otherwise, let n be the maximum value that is contained in $T(i,j)$ for some $(i,j) \in I(w)$. Choose i and j such that $T(i,j) = n$ and $i + j$ is as small as possible. Necessarily, $j = i + 1$, or in other words i is a right descent of w. If $|T(i,i+1)| > 1$, remove n from $T(i,i+1)$ to obtain a new function $T' : I(w) \to \mathbf{2}^{[n]} \setminus \{\varnothing\}$. If $|T(i,i+1)| = 1$, then let $T' : I(ws_i) \to \mathbf{2}^{[n]} \setminus \{\varnothing\}$ be the function defined by

$$T'(a,b) = T(s_i(a), s_i(b))$$

for all $(a,b) \in I(ws_i)$.

It is straightforward to check that T' is a set-valued inversions tableau, either for w if $|T(i,i+1)| > 1$ or for ws_n if $|T(i,i+1)| = 1$. Let $G' = \Phi(T')$. Then let $\Phi(T) = G' \cup \{(n, i+1-n)\}$.

Proposition 2.0.3. *Φ is a bijection between set-valued inversions tableaux of w with maximum value n and staircase word graphs G such that $w_G = w$ and $a \leq n$ for all $(a,b) \in G$.*

Proof. If $|T(i,j)| = 1$ for all $(i,j) \in I(w)$, then T is just an ordinary inversion tableau and $\Phi(T)$ is just the usual RC graph associated to T. In this case, the result is [1]. Otherwise, we prove this by induction on the maximum size of $|T(i,j)|$. Suppose this maximum size is $m > 1$ and we have proved the result for all set-valued inversions tableaux with maximum size less than m. We may assume without loss of generality that m is the maximum value that is contained in $T(i,j)$ for some $(i,j) \in I(w)$ by removing elements below.

In that case, $m \in T(i, i+1)$ for some i. Let T' be the set-valued inversions tableau obtained by removing m from $T(i, i+1)$ if $|T(i, i+1)| > 1$ or by applying the transformation described in the definition of Φ if $|T(i, i+1)| = 1$. By the inductive hypothesis, $\Phi(T')$ is a staircase word graph such that $w_{\Phi(T')} = w$ or $w_{\Phi(T')} = ws_i$ and $a \leq n$ for all $(a, b) \in \Phi(T')$. In either case, we have $w_{\Phi(T)} = w$. It is also clear that $a \leq n$ for all $(a, b) \in \Phi(T)$. This proves that $\Phi(T)$ is a staircase word graph with the desired properties.

Let G be a staircase word graph. Define $\Psi(G)$ to be empty if G is empty. Otherwise, let (i, j) be the maximum element of G and let $G' = G \setminus \{(i, j)\}$. Let $w' = w_{G'}$. Let $d = i + j - 1$. If $\ell(w's_d) > \ell(w')$, then let

$$\Psi(G)(a, b) = \begin{cases} \Psi(G')(s_d(a), s_d(b)) & \text{if } (a, b) \in I(w) \setminus \{(i, j)\} \\ \{d\} & \text{if } (a, b) = (i, j) \end{cases}$$

If $\ell(w's_d) < \ell(w')$, then let

$$\Psi(G)(a, b) = \begin{cases} \Psi(G')(a, b) & \text{if } (a, b) \in I(w) \setminus \{(i, j)\} \\ \Psi(G')(a, b) \cup \{d\} & \text{if } (a, b) = (i, j) \end{cases}$$

$\square$

Definition 2.0.4. Define the isobaric divided difference operator

$$\pi^i = \partial^i(1 + \beta x_{i+1})$$

Note the Leibniz formula

$$\pi^i(fg) = (\pi^i + \beta s_i)(f)g + f\pi^i(g)$$

Definition 2.0.5 (Pulling out an empty row)**.** Let R be a bounded RC graph with $\mathtt{ht}(R) = n$ and let $p \leq n$ be such that row p of R is empty. Let $R_{\leq p}$ denote the RC-graph consisting of the first p rows and $R_{>p}$ the RC-graph consisting of rows $p+1$ through n. Now, $\mathfrak{m}(R_{\leq p})$ could be greater than p. If it is less than or equal to p, define

$$\mathcal{Z}^{(p)}(R) = \mathcal{Z}(R_{\leq p}) \cup \downarrow R_{>p}$$

as a bounded RC graph with $\mathtt{ht}(R) - 1$ rows. Otherwise, define $R' = R_{\leq p}$, let $d_1 = \mathfrak{m}(w_{R'})$ and set $R_1 = R'$. While $\mathfrak{m}(R_i) > p$, do:

Set $R' = R_i$. While $\mathfrak{m}(R') \geq d_i$, delete the element $(a, b) \in R'$ such that $\mathtt{rt}_{R'}(a, b) = (\mathfrak{m}(R'), \mathfrak{m}(R') + 1)$ from R' and store a in the list (multiset) A_i. After enough iterations such that $\mathfrak{m}(R') < d_i$, set $R_{i+1} = R'$ and set $d_{i+1} = \mathfrak{m}(R_{i+1})$, then repeat.

At termination, say at step k, set

$$\mathcal{Z}^{(p)}(R) = (A_1 \overset{d_1-1}{\rightsquigarrow} \cdots A_k \overset{d_k-1}{\rightsquigarrow} \mathcal{Z}(\overline{R})) \cup \downarrow R_{>p}$$

be the desired bounded RC graph with $\mathtt{ht}(R) - 1$ rows.

We show an example in Figure 1.

Definition 2.0.6. Let R be an RC graph. We define an RC graph $\overline{R}$ and a sequence $\mathtt{strip}(R)$ as follows. Let $\mathfrak{m}(R) = n$, set $R_0 = R$, and set $A_0 = ()$. While $\mathfrak{m}(R_i) > n - 1$, delete the element $(a, b) \in R_i$ such that $\mathtt{rt}_{R_i}(a, b) = (\mathfrak{m}(R_i), \mathfrak{m}(R_i) + 1)$ to obtain R_{i+1}, and add a to A_i to obtain A_{i+1}. When the algorithm terminates at step k, set

$$\mathtt{strip}(R) = A_k$$

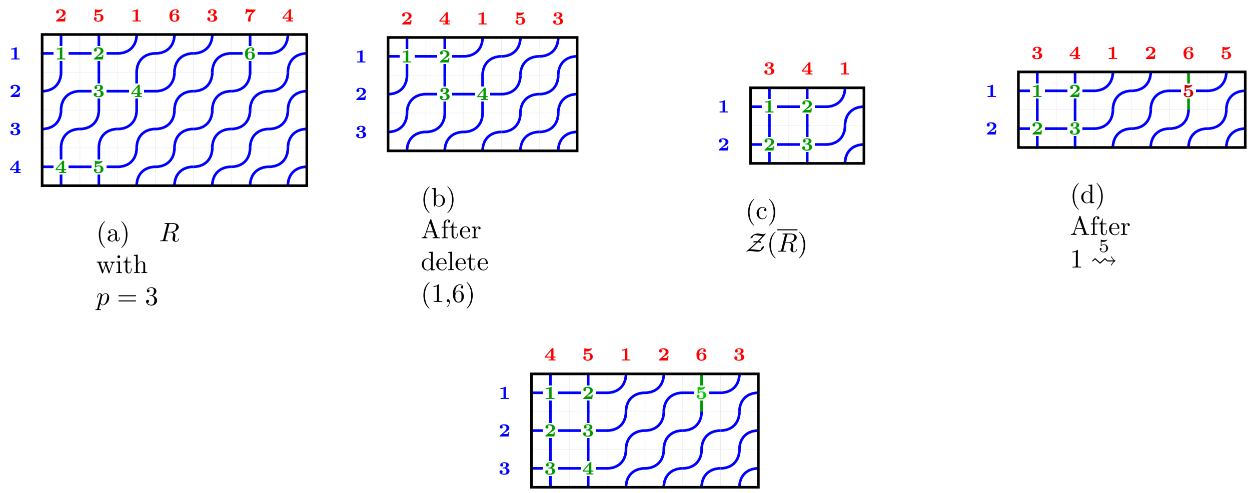
and set $\overline{R} = \overline{R}$.

Lemma 2.0.7. *Let R be an RC graph. Then*

$$R = \mathtt{strip}(R) \overset{\mathfrak{m}(R)}{\rightsquigarrow} \overline{R}$$

Proof. Let $n = \mathfrak{m}(R)$. We prove this by induction on $\mathfrak{c}_n(R)$. Let (i, j) be the location of the descent n in R. If $\mathfrak{c}_n(R) = 1$, then we know that insertion of i into $\overline{R}$ returns an RC graph with the same weight and permutation as R, the latter assertion because the only n-reflection that increases the length of $w_{\overline{R}}$ by exactly 1 is s_n. By the exchange property, there is precisely one location in row i that this n-reflection can be inserted, and we have the result when $\mathfrak{c}_n(R) = 1$.

Assume the inductive hypothesis that we have the result for $\mathfrak{c}_n(R) = k$ for some $k \geq 1$ and additionally that insertion order of reflections has been $(n, n+1), \ldots, (n, n+k)$. Assume $\mathtt{strip}(R) = (r_1, \ldots, r_{k+1})$ in



(a) R with $p=3$

(b) After delete (1,6)

(c) $\mathcal{Z}(\overline{R})$

(d) After $1 \overset{5}{\rightsquigarrow}$

(e) Final $\mathcal{Z}^{(p)}(R)$ with $R_{>p}$ reattached

FIGURE 1. Example of $\mathcal{Z}^{(p)}$ operation with $p=3$. Starting with (a) R where row 3 is empty, we identify element $(1,6)$ creates maxd $> p$. (b) After deleting it (storing row 1 in A_1). (c) Apply $\mathcal{Z}$ to get 2-row graph. (d) Insert back at row 1 with $k = d_1 - 1 = 5$ (shown in red). (e) Reattach $R_{>p}$ shifted down to get final result.

increasing order and we have inserted all of $r_2, \ldots, r_{k+1}$ in Algorithm **??**. By the induction hypothesis, the RC graph R' we have at this point has permutation $\mathfrak{c}_i R' = \mathfrak{c}_i R$ for $i < n$ and $\mathfrak{c}_n(R') = k$, and we have equality of R' with R except deletion of the root $(n, n+k+1)$. Since $w_{R'}(n) < w_{R'}(n+k+1)$, the only n-reflection that can be inserted to increase the length of $w_{R'}$ by 1 is $(n, n+k+1)$. By pipe considerations, this must be inserted in column $c = n+k+1-r_1$. Since this is the only possible insertion location and the permutation is the same as w_R, we have the result by induction. □

Theorem 2.2. *Let $R \in \mathcal{BRC}(w,n)^{(0,p)}$ be the set of all bounded RC graphs with $\mathtt{ht}(R) = n$ such that row p of R is empty. Then we have that the restriction of $\mathcal{Z}^{(p)}$ is a bijection*

$$\mathcal{Z}^{(p)} : \mathcal{BRC}(w,n)^{(0,p)} \to \bigcup_{\substack{w \overset{p}{\searrow} w' \\ \ell(w)=\ell(w')}} \mathcal{BRC}(w', n-1)$$

satisfying $\mathtt{wt}(\mathcal{Z}^{(p)}(R)) = (i_1, \ldots, i_{p-1}, i_{p+1}, \ldots, i_n)$, where $\mathtt{wt}(R) = (i_1, \ldots, i_n)$.

Proof. Consider first the special case that $R_{>p}$ is empty. Let $w_0 = w_{\mathcal{Z}(\overline{R})}$. Then we know that

$$\overline{R} \overset{p}{\searrow} \mathcal{Z}(\overline{R})$$

Since every element of A_k is less than p and $d_k > p$, the insertion $A_k \overset{d_k-1}{\rightsquigarrow} \mathcal{Z}(\overline{R})$ only adds right roots (a,b) with $a < p$ and $b > p$. Thus

$$A_k \overset{d_k}{\rightsquigarrow} \overline{R} \overset{p}{\searrow} A_k \overset{d_k-1}{\rightsquigarrow} \mathcal{Z}(\overline{R})$$

We assert the (descending) inductive hypothesis that for some index j we have

$$A_j \overset{d_j}{\rightsquigarrow} \cdots A_k \overset{d_k}{\rightsquigarrow} \overline{R} \overset{p}{\searrow} A_j \overset{d_j-1}{\rightsquigarrow} \cdots A_k \overset{d_k-1}{\rightsquigarrow} \mathcal{Z}(\overline{R})$$

Then the induction step follows with precisely the same reasoning; since every element of A_{j-1} is less than p and $d_{j-1} > p$, the insertion $A_{j-1} \overset{d_{j-1}-1}{\rightsquigarrow}$ only adds right roots (a,b) with $a < p$ and $b > p$. Thus

$$A_{j-1} \overset{d_{j-1}}{\rightsquigarrow} \cdots A_k \overset{d_k}{\rightsquigarrow} \overline{R} \overset{p}{\searrow} A_{j-1} \overset{d_{j-1}-1}{\rightsquigarrow} \cdots A_k \overset{d_k-1}{\rightsquigarrow} \mathcal{Z}(\overline{R})$$

and the result follows by induction in the special case that $R_{>p}$ is empty, where we invoke Lemma 2.0.7 to assert that the left-hand side is equal to R.

Since the row was initially empty, we have at our disposal the exchange property on $R_{>p}$, shifting the reflections down a row to obtain the general case. $\square$

[2]

References

[1] Axelrod-Freed, I. Inversions tableaux. *arXiv preprint arXiv:2507.11516* (2025).

[2] Samuel, M. J. A Molev-Sagan type formula for double Schubert polynomials. *J. Pure Appl. Algebra 228*, 7 (2024), 107636.